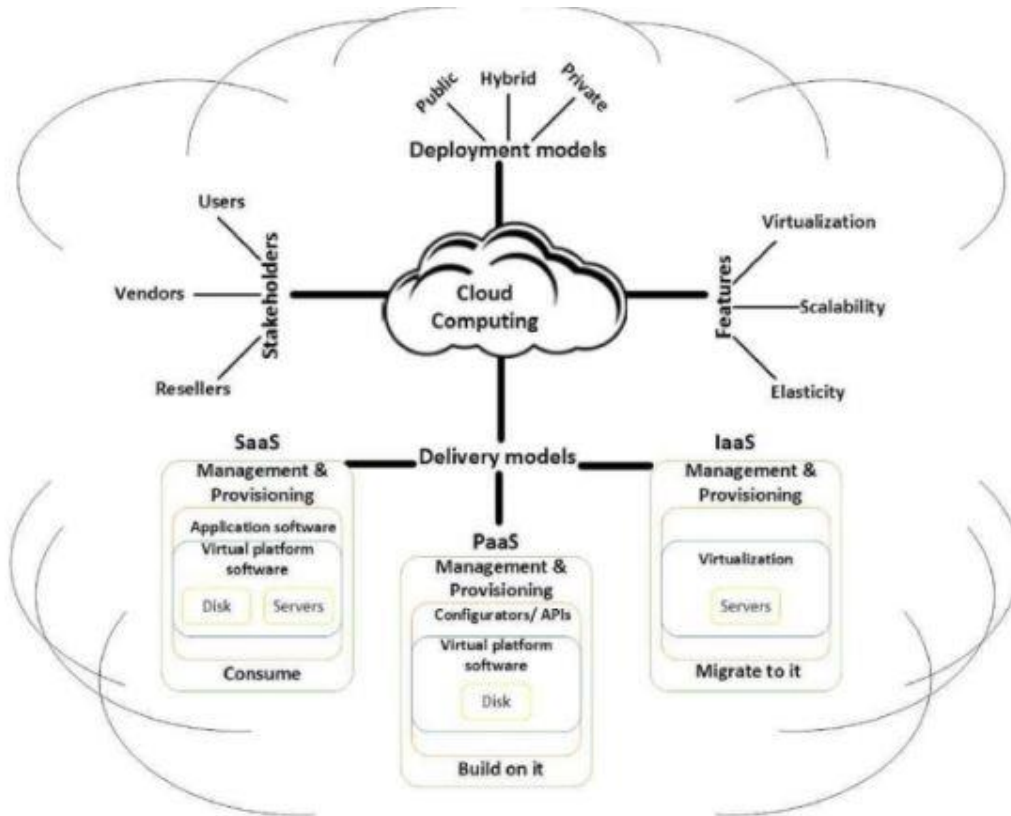


Experiment – 1

1) Aim/Objective:

Study of Cloud Service Models

2) Theory:



1. Infrastructure as a Service (IaaS)

Definition:

IaaS provides virtualized computing resources over the internet, such as virtual machines, storage, and networking infrastructure.

Key Features:

- Full control over hardware resources.
- Scalability to meet varying demands.
- Pay-as-you-go pricing for resources.

Examples:

- Amazon EC2
- Google Compute Engine
- Microsoft Azure Virtual Machines

Use Cases:

- Hosting websites or web applications.
- Setting up and managing virtual data centers.
- Running high-performance computing (HPC) applications.

Advantages:

- Flexible and customizable.
- No upfront hardware costs.
- Ideal for businesses needing control over infrastructure.

Challenges:

- Requires technical expertise for setup and management.
- Responsibility for maintaining the operating system and applications.

2. Platform as a Service (PaaS)

Definition:

PaaS provides a platform for developers to build, deploy, and manage applications without dealing with underlying infrastructure.

Key Features:

- Pre-configured environments for application development.
- Integrated tools for testing and deployment.
- Managed services for scalability.

Examples:

- Microsoft Azure App Service
- AWS Elastic Beanstalk

Use Cases:

- Developing mobile or web applications.
- Prototyping and rapid application development.
- Automating application deployment and scaling.

Advantages:

- Simplifies the development process.
- Reduces the need for infrastructure management.
- Supports multiple programming languages and frameworks.

Challenges:

- Limited control over infrastructure and runtime environments.
- Dependency on provider-specific configurations.

3. Software as a Service (SaaS)

Definition:

SaaS delivers software applications over the internet on a subscription basis, eliminating the need for installation and maintenance.

Key Features:

- Accessible via web browsers or APIs.
- Fully managed by the service provider.
- Typically involves a subscription or usage-based pricing model.

Examples:

- Google Workspace (Docs, Sheets, Gmail)
- Microsoft 365
- Salesforce

Use Cases:

- Email and collaboration tools.
- Customer relationship management (CRM) software.
- File storage and sharing solutions.

Advantages:

- Easy to use with minimal setup.
- Regular updates and maintenance handled by the provider.
- Cost-effective for businesses without technical expertise.

Challenges:

- 1) Limited customization options.
- 2) Reliance on internet connectivity and vendor reliability.

Comparison Table

Aspect	IaaS	PaaS	SaaS
Control	High (infrastructure-level)	Medium (platform-level)	Low (application-level)
Management	By customer	Shared (platform managed, code by customer)	By provider
Flexibility	Maximum	Moderate	Minimum
Technical Expertise	High	Moderate	Low
Use Case	Custom IT solutions	Application development	End-user software

Conclusion

- **IaaS** is suitable for businesses requiring flexibility and control over infrastructure.
- **PaaS** is ideal for developers who want to focus on application development without managing servers.
- **SaaS** is perfect for businesses or individuals needing ready-to-use software with minimal maintenance responsibilities.

Viva Questions:

- 1) What is cloud computing?
- 2) Name the three cloud service models.
- 3) Differentiate public and private cloud.